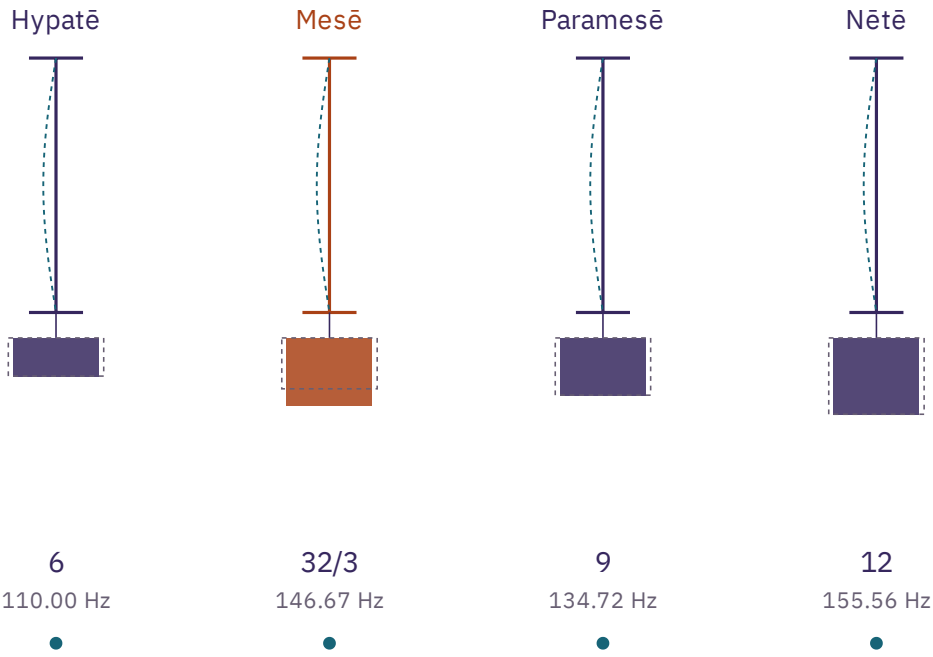
Double the weight. Half the octave.

Nicomachus's *Harmonic Manual*, chapter 6, calls the equal-length strings loaded with 12 and 6 an octave. A modern ideal string model gives √2 as the frequency ratio: 600 cents, half of the claimed 1200-cent interval.

Your experiment: fixed load 6 at 110 Hz; selected string 2. Original source numbers remain 6, 8, 9, 12. The reference pitch is modern and arbitrary.

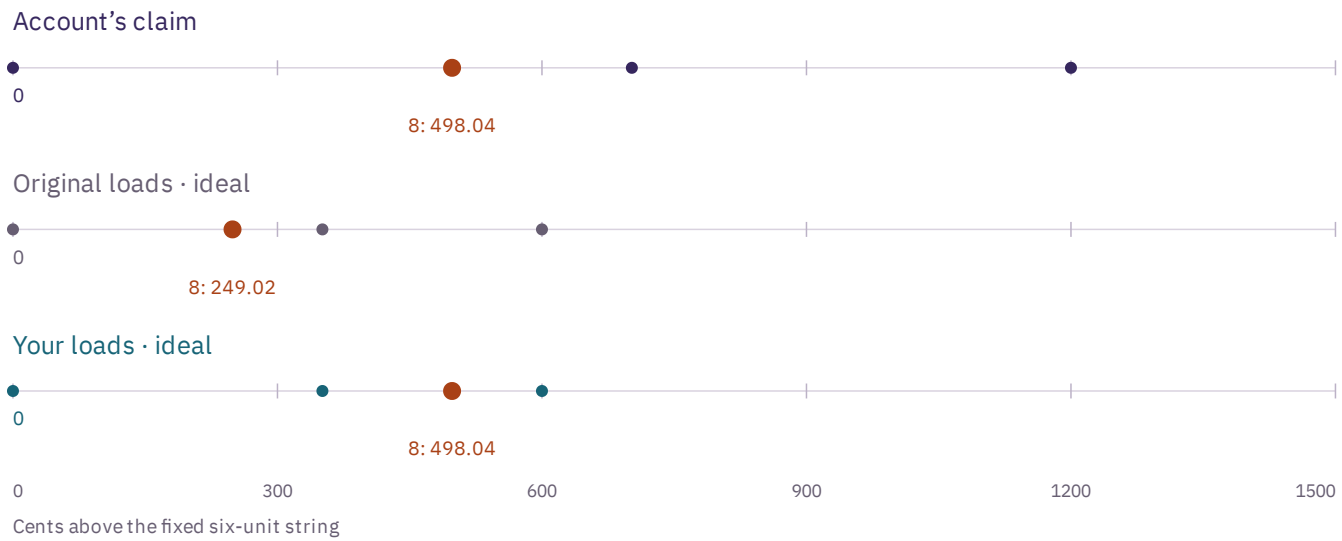


Solid rectangles: your relative loads. Hollow outlines: original loads. All vibrating lengths stay equal. Rectangle height encodes load; it does not reconstruct the shape of a weight. The dashed first-mode outline has arbitrary amplitude. The account uses one shared attachment; these columns are a comparison.

Hear the distinction

Selected target and your model together. In the comparison sequence: claimed low/high pair, original ideal low/high pair, then your ideal low/high pair. The four-note choices use the source's numerical skeleton, not a full scale. The simultaneous choice sounds the selected target with your ideal note.

Generated sine fundamentals. No recovered ancient pitch, melody, performance, hammer sound or instrumental timbre.



A fixed cents ruler. Original numbers label the notes, even in the row with your replacement loads. The octave at 1200 cents stays in the same place.

All ratios compare with the fixed load 6 and 110 Hz reference. Irrational frequencies are rounded, not exact fractions.

Note	Source load	Claimed f/f_0	Original ideal f/f_0	Your load	Your ideal Hz	From target	Exact repair load
Hypatē	6	1	1	6	110.000000	0.000000 cents	6
Mesē	8	4/3	√(4/3)	32/3	146.666667	0.000000 cents	32/3
Paramesē	9	3/2	√(3/2)	9	134.721936	-350.977500 cents	27/2
Nētē	12	2	√(2)	12	155.563492	-600.000000 cents	24

Why the weights must be squared

For an ideal string, $f = (1 / 2L)\sqrt{(T/\mu)}$. With loaded vibrating length L and linear mass density μ held fixed, and tension T proportional to load W, $f/f_0 = \sqrt{(W/6)}$. To reach a target frequency ratio r, solve $W = 6r^2$.

Thus 4/3 requires 32/3 units; 3/2 requires 27/2; 2 requires 24. These are modern repairs of the model, not emendations of the ancient text. With the original loads, every interval in cents is halved because $\log_2\sqrt{r} = \frac{1}{2}\log_2 r$. An octave is 1200 cents, a modern logarithmic measure.

Modern ideal comparison: uniform flexible strings, fixed vibrating length and equal mass per length after loading, constant tension proportional to load, small transverse vibration and fixed endpoints. String mass compared with load, elastic stretching, changing density, friction, stiffness, support motion and damping are omitted. The diagram is a comparison, not a fixture reconstruction; the sound is synthetic sine fundamentals, not hammer timbre or an ancient performance.

These computed comparisons test stated conditions within an ideal model; no physical instrument was measured. Changing vibrating length would be a different experiment. OpenStax's force derivation gives the modern wave law; Harvard's sonometer demonstration separately describes the load comparison.

Read the account, then question its inference

The Greek excerpts follow Jan's 1895 critical text, chapter 6, pp.245–248. The Latin is Meibom's own 1652 translation, Nicomachus pp.10–13. They are separate historical editions. Greek ligatures and line breaks, Latin long-s and ae/oe ligatures, spacing and punctuation are normalized. Ellipses mark internal omissions. English translations and grammar notes are new. This is a later literary account of Pythagoras, not a surviving laboratory report by him. Its hammer story is retained in the full pages; the model here concerns only the explicitly weighted strings.

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1. Four strings, four conditions

ἀπαρτήσας τέσσαρας χορδὰς ὁμοΰλους καὶ ἰσοκώλους, ἰσοπαχεῖς τε καὶ ἰσοστρόφους

adpendens quatuor chordas ejusdem materiae existentes, tum aeque longas, & aeque crassas, atque aeque graves

Having suspended four strings of the same material and equal length, equal in thickness and equally twisted…

ἀπαρτήσας is an aorist active participle: "having suspended". The four accusative plural adjectives describe χορδάς. ἰσοστρόφους says "equally twisted"; Meibom's aeque graves says "equally heavy". Keep that translation difference visible. The ideal model assumes equal mass per vibrating length after loading. Greek: jan-247; Latin: meibom-11.

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2. Length is deliberately held equal

τὰ δὲ μήκη τῶν χορδῶν μηχανησάμενος ἐκ παντὸς ἰσαίτατα

Cumque ita chordarum longitudines omnino aequales effecisset

And having contrived to make the lengths of the strings entirely equal…

μήκη is the accusative plural of μῆκος, "length"; μηχανησάμενος is an aorist middle participle, "having arranged/contrived". The repeated length condition matters: the comparison changes load, not vibrating length. Latin effecisset is pluperfect subjunctive after cum. Greek: jan-247; Latin: meibom-11.

3. The largest and smallest load

τὴν μὲν γὰρ ὑπὸ τοῦ μεγίστου ἐξαρτήματος τεινομένην πρὸς τὴν ὑπὸ τοῦ μικροτάτου διὰ πασῶν φθεγγομένην κατελάμβανεν.

Namque a maximo pondere tensam, ad eam quae a minimo, dia pason sonantem deprehendit.

For he found the string stretched by the greatest hanging load sounding an octave against the one stretched by the smallest.

τεινομένην, a present passive participle of τείνω, describes the string being stretched. The repeated feminine article τήν supplies “string”. πρὸς marks comparison, not a direction of motion. The Latin sentence crosses printed pp.11–12; both complete pages are retained. Greek: jan-247; Latin: meibom-12.

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4. Twelve and six, called double

ἦν δὲ ἡ μὲν δώδεκά τινων ὁλκῶν, ἡ δὲ ἕξ. ἐν διπλασίῳ δὴ λόγῳ ἀπέφαινε τὴν διὰ πασῶν, ὅπερ καὶ αὐτὰ τὰ βάρη ὑπέφαινε.

Erat autem illa duodecim quarundam librarum; haec vero sex. Atque ita in dupla ratione constituebat dia pason consonantiam, quam & ipsae gravitates ostendebant.

One was of twelve unspecified weight units, the other six. So he declared the octave to be in a double ratio, which the weights themselves also indicated.

μέν … δέ balances the two quantities. λόγῳ is dative singular of λόγος, here “ratio”. ὁλκῶν specifies units of weight; τινων leaves them unspecified. Meibom chooses librarum. This lesson uses relative load units, without identifying an ancient pound or converting it into kilograms. Greek: jan-247; Latin: meibom-12.

5. Twelve against eight, a fifth

τὴν δ’ αὖ μεγίστην πρὸς τὴν παρὰ τὴν μικροτάτην (οὖσαν ὀκτὼ ὁλκῶν) διὰ πέντε συμφωνοῦσαν, ἔνθεν ταύτην ἀπέφαινεν ἐν ἡμιολίῳ λόγῳ

Rursus maximam ad juxta minimam, quae octo librarum existebat, dia pente consonantem invenit. unde hanc in ratione sesquialtera constituit

Again, the greatest against the one next to the smallest (of eight weight units) sounded a fifth; from this he assigned it the ratio of one and a half.

ἡμιόλιος and Latin sesquialter describe 3:2: one whole and its half. At fixed length and linear density, a 3:2 load ratio produces √(3/2) as a frequency ratio. The ideal interval in cents is half the claimed fifth. Greek: jan-247; Latin: meibom-12.

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6. Nine against eight, a tone

τὸ ἄρα μεταξὺ τῆς διὰ πέντε καὶ τῆς διὰ τεσσάρων τουτέστιν ᾧ ὑπερέχει ἡ διὰ πέντε τῆς διὰ τεσσάρων, ἐβεβαιοῦτο ἐν ἐπογδόῳ λόγῳ ὑπάρχειν, ἐν ᾧπερ τὰ ἐννέα πρὸς τὰ ὀκτώ.

Quod itaque est inter dia pente & dia tessaron, hoc est, quo dia pente consonantia superat dia tessaron, confirmatum est in superoctava esse ratione, in qua novem ad octo.

Thus the difference between the fifth and the fourth—that by which the fifth exceeds the fourth—was established as the ratio of one and an eighth, in which nine stands to eight.

ἐπόγδοος and superoctavus mean a whole plus its eighth, 9:8. The interval argument is numerical: (3/2) ÷ (4/3) = 9/8. That arithmetic remains valid even though the stated string loads do not produce the claimed frequency ratios under the ideal law. Greek: jan-247; Latin: meibom-12.

7. The story says there was one attachment

τὴν μὲν τῶν χορδῶν κοινὴν ἀπόδεσιν τὴν ἐκ τοῦ διαγωνίου πασσάλου

communem quidem chordarum adpensionem, quae ex palo erat ab angulo ad angulum infixo

The common attachment of the strings from the diagonal peg…

κοινήν means “common/shared”; πασσάλου is genitive, “of the peg”. This later phrase recalls the single peg introduced on p.246. Our four separate columns compare equal lengths; they do not reconstruct the ancient fixture. The modern ideal model treats loaded string tension as proportional to the hanging weight and uses fixed vibrating endpoints. Greek: jan-248; Latin: meibom-13.

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8. Four named pitches, not a complete scale

ὑπάτην μὲν τὸν τοῦ ἓξ ἀριθμοῦ κοινωνοῦντα φθόγγον, μέσην δὲ τὸν τοῦ ὀκτὼ … παραμέσην δὲ τὸν τοῦ ἐννέα … νήτην δὲ τὸν τοῦ δώδεκα

hypaten, numeri senarii participem sonum: mesen, octonarii … paramesen, novenarii participem … neten denique, duodenarii

Hypatē, the note associated with six; mesē, with eight … paramesē, with nine … and nētē, with twelve.

φθόγγον is the object “note/sound”. The names attach to the four source numbers. Ellipses here omit descriptions of their intervals; the page then says that the intervening diatonic notes were supplied. Our four tones are this numerical skeleton, not a recovered tune or the full eight-note system. Greek: jan-248; Latin: meibom-13.

Words for a controlled comparison

χορδάς · chordās · strings

Accusative plural of χορδή; the object of “having suspended”. Latin chordas.

ἰσοκώλους · isokōlous · equal in length

ισο- (equal) + κῶλον (limb/member); one of the four adjectives agreeing with strings.

ἰσοπαχεῖς · isopacheis · equal in thickness

Jan’s main text. His apparatus records ισοπαγεῖς in M and B; the complete scan preserves the variant.

ἰσοστρόφους · isostrophous · equally twisted

ἰσο- + στρέφω (turn/twist). Compare the printed Latin aeque graves, “equally heavy”.

μήκη · mēkē · lengths

Accusative plural of μῆκος. The explicit repeated condition is crucial to the physical comparison.

τεινομένην · teinomenēn · being stretched

Present passive participle, feminine accusative, agreeing with the understood string. Latin tensam.

λόγῳ · logōi · in a ratio

Dative of λόγος; in this context a ratio, rather than speech or an argument. Latin ratione.

ἡμιολίῳ · hēmioliōi · in a 3:2 ratio

A whole plus its half. Latin sesquialtera, an adjective agreeing with ratione.

ἐπιτρίτῳ · epitritōi · in a 4:3 ratio

A whole plus its third. Latin supertertia; the adjacent source also uses superquarta when describing 9 against 12.

ἐπογδόῳ · epogdoōi · in a 9:8 ratio

A whole plus its eighth. Latin superoctava. The source’s musical tone is 9:8.

Reopen this experiment · Source and model decisions.

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ὄγδοόν τινα φθόγγον μεταξὺ μέσης καὶ παραμέσης ἐνάψας καὶ ἀποστήσας ἀπὸ μὲν τῆς μέσης ὅλον τόνον, ἀπὸ δὲ τῆς παραμέσης ἡμιτόνιον· ὥστε τὴν μὲν προτέραν ἐν τῇ ἑπταχόρδῳ παραμέσην οὖσαν τρίτην ἔτι ἀπὸ νήτης καλεῖσθαί τε καὶ οὐδὲν ἧττον κεῖσθαι, τὴν δὲ παρεντεθεῖσαν (p. 10) τετάρτην μὲν ἀπὸ τῆς νήτης ὑπάρχειν, συμφωνεῖν δὲ πρὸς αὐτὴν τὴν διὰ τεσσάρων συμφωνίαν, ἥνπερ καὶ ἡ ἐξ ἀρχῆς μέση πρὸς τὴν ὑπάτην εἶχεν. ὁ δὲ μεταξὺ ἀμφοτέρων τόνος μέσης τε καὶ παρεντεθείσης, ὀνομασθείσης δὲ ἀντὶ τῆς προτέρας παραμέσης, ὁποτέρῳ ἂν τετραχόρδῳ προστεθῇ, εἴτε τῷ πρὸς τῇ ὑπάτῃ νητοειδέστερος, εἴτε τῷ πρὸς τῇ νήτῃ βομβυκέστερος, τὴν διὰ πέντε συμφωνίαν ἀποδείξει, σύστημα ἑκατέρων ὑπάρχουσαν αὐτοῦ τε τοῦ τετραχόρδου καὶ τοῦ προσγενομένου τόνου. ὥσπερ καὶ ὁ τῆς διὰ πέντε λόγος ὁ ἡμιόλιος σύστημα εὑρίσκεται ἐπιτρίτου τε ἅμα καὶ ἐπογδόου· ὁ ἄρα τόνος ἐπόγδοον.

6. *Πῶς οἱ ἀριθμητικοὶ τῶν φθόγγων λόγοι ηὑρέθησαν.*

ϛ. Τὴν δὲ κατ' ἀριθμὸν ποσότητα ταύτην ἥτε διὰ τεσσάρων χορδῶν ἀπόστασις ἥτε διὰ πέντε καὶ ἡ κατ' ἀμφοτέρων σύνοδον διὰ πασῶν λεγομένη καὶ ὁ προσκείμενος μεταξὺ τῶν δύο τετραχόρδων τόνος τρόπῳ τινὶ τοιούτῳ ὑπὸ τοῦ Πυθαγόρου καταληφθέντι ἔχειν ἐβεβαιοῦτο. ἐν φροντίδι ποτὲ καὶ διαλογισμῷ

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3 παρα R sup. lin. 4 τῇ MB, exc. Neap., τῷ R. 6 τῆς om. MB. 8 ἡ om. B. 10 δὲ om. M. 12 τῷ om. MB. 13 βομβυδέστ. R. 14 ἑκατέρων R (ν in ras.), ἑκατέρωσε MB, ἑκατέρωθεν exc. Neap. 16 τῆς RB, om. M. ὁ om. RB. 17 ἐπόγδοον RM¹, ἐπόγδοος B. 21 ὁ om. M. 23 καταληφθέντα M (et B?). 24 ἐβεβαιοῦτο R, ἐβεβαιώθη MB.

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συντεταμένῳ ὑπάρχων, εἰ ἄρα δύναιτο τῇ ἀκοῇ βοηθειάν τινα ὀργανικὴν ἐπινοῆσαι παγίαν καὶ ἀπαραλόγιστον, οἵαν ἡ μὲν ὄψις διὰ τοῦ διαβήτου καὶ διὰ τοῦ κανόνος ἢ καὶ διὰ τῆς διόπτρας ἔχει, ἡ δ' ἁφὴ διὰ τοῦ ζυγοῦ ἢ διὰ τῆς τῶν μέτρων ἐπινοίας, παρά τι χαλκοτυπεῖον περιπατῶν ἔκ τινος δαιμονίου συντυχίας ἐπήκουσε ῥαιστήρων σίδηρον ἐπ' ἄκμονι ῥαιόντων καὶ τοὺς ἤχους παραμὶξ πρὸς (p. 11) ἀλλήλους συμφωνοτάτους ἀποδιδόντων πλὴν μιᾶς συζυγίας· ἐπεγίνωσκε δ' ἐν αὐτοῖς τὴν δὲ διὰ πασῶν καὶ τὴν διὰ πέντε καὶ τὴν διὰ τεσσάρων συνῳδίαν. τὴν δὲ μεταξύτητα τῆς τε διὰ τεσσάρων καὶ τῆς διὰ πέντε ἀσύμφωνον μὲν ἑώρα αὐτὴν καθ' ἑαυτήν, συμπληρωτικὴν δὲ ἄλλως τῆς ἐν αὐτοῖς μείζονος. ἄσμενος δὴ ὡς κατὰ θεὸν ἀνυομένης αὐτῷ τῆς προθέσεως εἰσέδραμεν εἰς τὸ χαλκεῖον καὶ ποικίλαις πείραις παρὰ τὸν ἐν τοῖς ῥαιστῆρσιν ὄγκον εὑρὼν τὴν διαφορὰν τοῦ ἤχου, ἀλλ' οὐ παρὰ τὴν τῶν ῥαιόντων βίαν οὐδὲ παρὰ τὰ σχήματα τῶν σφυρῶν οὐδὲ παρὰ τὴν τοῦ ἐλαυνομένου σιδήρου μετάθεσιν, σηκώματα ἀκριβῶς ἐκλαβὼν καὶ ῥοπὰς ἰσαιτάτας τῶν ῥαιστήρων πρὸς ἑαυτὸν ἀπηλλάγη. καὶ ἀπό τινος ἑνὸς πασσάλου διὰ γώνων ἐμπεπηγότος τοῖς τοίχοις, ἵνα μὴ κἀκ τούτου διαφορά τις ὑποφαίνηται ἢ ὅλως ὑπονοῆται πασσάλων ἰδιαζόντων παραλ-

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1 *συντεταμένῳ* Iamblich vita Pyth. c. 26, ita corr. vid. R, *συντεταγμένων* MR¹, *συντεταγμένῳ* B. 2 *ὀργανικὴν* om. M. 4 *καὶ διὰ τῆς* ego, *νὴ δία διὰ* (B add. *τῆς*) libb. et Iambl., *ἔχει* Iambl., *ἔχεν* RB, *ἔσχεν* M. 5 *τοῦ* om. M (hab. Iambl.) *ἢ δία* MB Iambl., *ἢ καὶ δία* R. 6 *χαλκοτύπιον* M solus. 7 *σίδηρον* MIambl., *σιδηρῶν* RB. 15 *ἀμυνομένης* R. 16 *τὸν* et *ὄγκον* R, *τῶν* et *ὄγκων* MBIambl. 20 *ἐλαβών* R, *ἐκβαλών* M. 22 *διὰ γώνων* R, *διαγων* M¹, *διαγωνίου* M²B. 23 *ἀποφαίνηται* R.

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λαγή, ἀπαρτήσας τέσσαρας χορδὰς ὁμοΰλους καὶ ἰσοκώλους, ἰσοπαχεῖς τε καὶ ἰσοστρόφους ἑκάστην ἐφ' ἑκάστης ἐξήρτησεν, ὁλκὴν προσδήσας ἐκ τοῦ κάτωθεν μέρους. τὰ δὲ μήκη τῶν χορδῶν μηχανησάμενος ἐκ παντὸς ἰσαίτατα, εἶτα κρούων ἀνὰ δύο ἅμα χορδὰς ἐναλλὰξ συμφωνίας εὕρισκε τὰς προλεχθείσας, ἄλλην ἐν ἄλλῃ συζυγίᾳ. τὴν μὲν γὰρ ὑπὸ τοῦ μεγίστου ἐξαρτήματος τεινομένην πρὸς τὴν ὑπὸ τοῦ μικροτάτου (p. 12) διὰ πασῶν φθεγγομένην κατελάμβανεν. ἦν δὲ ἡ μὲν δώδεκά τινων ὁλκῶν, ἡ δὲ ἕξ. ἐν διπλασίῳ δὴ λόγῳ ἀπέφαινε τὴν διὰ πασῶν, ὅπερ καὶ αὐτὰ τὰ βάρη ὑπέφαινε. τὴν δ' αὖ μεγίστην πρὸς τὴν παρὰ τὴν μικροτάτην (οὖσαν ὀκτὼ ὁλκῶν) διὰ πέντε συμφωνοῦσαν, ἔνθεν ταύτην ἀπέφαινεν ἐν ἡμιολίῳ λόγῳ, ἐν ᾧπερ καὶ αἱ ὁλκαὶ ὑπῆρχον πρὸς ἀλλήλας· πρὸς δὲ τὴν μεθ' ἑαυτὴν μὲν τῷ βάρει, τῶν δὲ λοιπῶν μείζονα, ἐννέα σταθμῶν ὑπάρχουσαν, τὴν διὰ τεσσάρων, ἀναλόγως τοῖς βρίθεσι. καὶ ταύτην δὴ ἐπίτριτον ἄντικρυς κατελαμβάνετο, ἡμιολίαν τὴν αὐτὴν φύσει ὑπάρχουσαν τῆς μικροτάτης, (τὰ γὰρ ἐννέα πρὸς τὰ ἕξ οὕτως ἔχει,) ὅνπερ τρόπον ἡ παρὰ τὴν μικρὰν ἡ ὀκτὼ πρὸς μὲν τὴν τὰ ἕξ ἔχουσαν ἐν ἐπιτρίτῳ ἦν, πρὸς δὲ τὴν τὰ δώδεκα ἐν ἡμιολίῳ. τὸ ἄρα μεταξὺ τῆς διὰ πέντε καὶ τῆς διὰ τεσσάρων τουτέστιν ᾧ ὑπερέχει ἡ διὰ πέντε τῆς διὰ τεσσάρων, ἐβεβαιοῦτο ἐν ἐπογδόῳ λόγῳ ὑπάρχειν, ἐν ᾧπερ τὰ ἐννέα πρὸς τὰ ὀκτώ. ἑκατέρως τε

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1 ἀπαρτίσας R. 2 ἰσοπαγεῖς MB. 11 τὰ om. B. 12 post ὑπέφαινε rep. τὴν διὰ πασῶν R. παρὰ τὴν μικρ. R Iambl. (cf. lin. 21), παραμικρ. MB. 18 ἐπίριτον ἄντικρυς i. e. 9 : 12. 24 ᾧ] ὥπερ R. 26 ἑκατέρως M nigro colore (M[1])? et Iambl., ἑκατέρων RB, M livido colore, Uen.

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ἡ διὰ πασῶν σύστημα ἠλέγχετο τῆς διὰ πέντε καὶ διὰ τεσσάρων ἐν συναφῇ, ὡς ὁ διπλάσιος λόγος ἥτοι ἡμιολίου τε καὶ ἐπιτρίτου, οἷον δώδεκα ὀκτὼ ἓξ, ἢ ἀναστρόφως τῆς διὰ τεσσάρων καὶ διὰ πέντε, ὡς τὸ διπλάσιον ἐπιτρίτου τε (p. 13) καὶ ἡμιολίου, οἷον δώδεκα ἐννέα ἓξ ἐν τάξει τοιαύτῃ. τυλώσας δὲ καὶ τὴν χεῖρα καὶ τὴν ἀκοὴν πρὸς τὰ ἐξαρτήματα καὶ βεβαιώσας πρὸς αὐτὰ τὸν τῶν σχέσεων λόγον, μετέθηκεν εὐμηχάνως τὴν μὲν τῶν χορδῶν κοινὴν ἀπόδεσιν τὴν ἐκ τοῦ διαγωνίου πασσάλου εἰς τὸν τοῦ ὀργάνου βατῆρα, ὃν χορδότονον ὠνόμαζε, τὴν δὲ ποσὴν ἐπίτασιν ἀναλόγως τοῖς βάρεσιν εἰς τὴν τῶν κολλάβων ἄνωθεν σύμμετρον περιστροφήν. ἐπιβάθρᾳ τε ταύτῃ χρώμενος καὶ οἷον ἀνεξαπατήτῳ γνώμονι εἰς ποικίλα ὄργανα τὴν πεῖραν λοιπὸν ἐξέτεινε, λεκίδων τε κροῦσιν καὶ αὐλοὺς καὶ σύριγγας καὶ μονόχορδα καὶ τρίγωνα καὶ τὰ παραπλήσια, καὶ σύμφωνον εὗρισκεν ἐν ἅπασι καὶ ἀπαράλλακτον τὴν δι' ἀριθμοῦ κατάληψιν. ὀνομάσας δὲ ὑπάτην μὲν τὸν τοῦ ἓξ ἀριθμοῦ κοινωνοῦντα φθόγγον, μέσην δὲ τὸν τοῦ ὀκτὼ, ἐπίτριτον αὐτοῦ τυγχάνοντα, παραμέσην δὲ τὸν τοῦ ἐννέα, τόνῳ τοῦ μέσου ὀξύτερον καὶ δὴ καὶ ἐπόγδοον, νήτην δὲ τὸν τοῦ δώδεκα, καὶ τὰς μεταξύτητας κατὰ τὸ διατονικὸν γένος συναναπληρώσας φθόγγοις ἀναλόγοις οὕτως τὴν ὀκτάχορδον ἀριθμοῖς συμφώνοις ὑπέταξε, διπλασίῳ ἡμιολίῳ ἐπιτρίτῳ καὶ τῇ τούτων διαφορᾷ ἐπογδόῳ.

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1 post ἠλέγχετο omnes ἥτοι, quod transposui post λόγος. 6 δὲ om. MB. 9 ἀπόδοσιν R. 11 χονδρότονον B. 12 κολάβων R. ἄνωθεν, erat enim chordotonon in inferiore, paxilli in superiore parte lyrae. 13 ἐπιστροφήν R. 14 τὴν π. λ. RIambl., λ. τὴν π. MB. 17 ἀπαράληπτον MB. 19 μὲν om. M. 24 συναπληρώσας M.

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à nete; ad illam verò consonet dia tessaron consonantiam, quam & ab initio mese ad hypaten habebat. Porro tonus qui inter ambas est, & mesen & interpositam; quæ prioris loco parameſe est appellata; utricunque tetrachordo adponatur: sive illi, quod est ad hypaten, ultimarum naturam magis referens; sive isti, quod ad neten, supremarum: dia pente consonantiam demonstrabit, systema utrinque existentem ipsius tetrachordi & adjuncti toni. quemadmodum & ipsius dia pente ratio sesquialtera, systema invenitur supertertiæ simul & superoctavæ. Vnde tonus est superoctavus.

Hanc autem secundùm numerum quantitatem; tum quæ per quatuor chordas distantia dicitur, tum quæ per quinque, tum quæ secundum utriusque concursum dia pason, & adjectus inter duo tetrachorda tonus; modo quodam tali à Pythagora deprehensam affirmant. Curis aliquando, & disquisitioni eorum quæ composuisset, intentus, num auditui aliquod excogitare posset adminiculum, firmum ac erroris expers: quale visus habet per normam, & per regulam, aut etiam per dioptram; tactus verò per libram, aut mensurarum excogitationem: prope æris officinam deambulans, divino quodam casu exaudiit malleos ferreos, super incude tundentes, sonitusque permiste quam maxime consonum, πλάρτιν μὲν ἀπὸ νήτης ὑπάρχειν· συμφωνεῖν δὲ πρὸς αὐτὴν τὴν διὰ πασῶν συμφωνίαν, ἥνπερ καὶ ἐξ ἀρχῆς μέση πρὸς τὴν ὑπάτην εἶχεν. Ὁ δὲ μεταξὺ ἀμφοτέρων τόνος, μέσης τε καὶ τῆς παρεντεθείσης, ὀνομαθείσης ἀνὶ τῆς προτέρας διὰ μέσης, ὁπότερῳ ἂν πλαγιαχόρδῳ προσπαθῇ· εἴτε τῷ πρὸς τὴν ὑπάτην, νηϊοειδέστερος, εἴτε πρὸς τὴν νήτην, βομβυκιέστερος, τὴν διὰ πέντε συμφωνίαν ἀποδείξει, σύστημα ἑκατέρωσε ὑπαρχουσαν, αὐτήν τε καὶ πλαγιαχόρδου, καὶ διὰ προσλαμβανόμενον τόνου. ὥσπερ καὶ ὁ διὰ πέντε λόγος, ὁ ἡμιόλιος, σύστημα διεζευγμένας ἐπιμερίτου τε ἅμα καὶ ἐπογδόου. ὁ ἄρα τόνος ἐπόγδοος.

Τοιοῦτον δὲ καὶ ἀριθμὸν ποσότητα φασὶν, ἥτε διὰ τεσσάρων χορδῶν ἀπόστασις, ἥτε διὰ πέντε, καὶ ἡ καθ' ἀμφοτέρων συῳδον, διὰ πασῶν λεγομένη. καὶ προσκείμενος μεταξὺ τῶν δύο πλαγιαχόρδων τόνος, τρόπῳ τινὶ τοιούτῳ ὑπὸ δὲ Πυθαγόρου καταληφθέντα ἔχειν ἐβεβαιώθη. Ἐν Φρυγίοις ποτὲ καὶ διαλογισμῷ σωτηριωδεστάτῳ ὑπάρχων, εἴ ἄρα δυναίατο τῇ ἀκοῇ βοήθειάν τινα ἐπινοῆσαι, παγίαν καὶ ἀπαραλόγιστον, οἵαν ἡ μὲν ὄψις διὰ δὲ διαβήτου, καὶ διὰ δὲ κανόνος, ἡ νὴ Δία διὰ διόπτρας ἔχει· ἡ δ' ἁφὴ διὰ ζυγοῦ, ἢ διὰ τοῦ τῶν μέτρων ἐπινοίας, ἀπό τινος χαλκοτύπου θείου παλῶν, ἔκ τινος δαιμονίου συντυχίας, ἐπήκουσε ῥαιστήρων σιδηρῶν ἐπ' ἄκμονι ῥαιόντων, καὶ τοὺς ἤχους ἁρμόζειν πρὸς ἀλλήλους

ἀλλήλας συμφωνοῦντας ἀποδιδόντων, πλὴν μιᾶς συζυγίας. ἐπεγίνωσκε δ' ἐν αὐτοῖς τήνδε διὰ παζῶν, καὶ τὴν διὰ πέντε, καὶ τὴν διὰ παςάρων συμφωνίαν. τὴν δὲ μεταξύ τινα τὴν διὰ παςάρων, καὶ τῆς διὰ πέντε, ἀσύμφωνον μὲν ἑώρα αὐτὴν καθ' ἑαυτήν· συμπληρωτικὴν δὲ ἄλλως τῆς ἐν αὐτοῖς μείζονος. ἄσμενος δὴ ὡς εἰκὸς ἦν ἀνυομένης αὐτῷ τῆς προθέσεως εἰς τὸ χαλκεῖον, καὶ πολλαῖς πείραις ἤδη ἐν τοῖς ῥαιστῆρσιν ὄγκους λίθων τὴν διαφορὰν ποιοῦντα. ἀλλ' οὐ διὰ τὴν τῶν ῥαιόντων βίαν. οὐδὲ διὰ τὰ σχήματα τῶν σφυρῶν, οὐδὲ διὰ τὴν τοῦ ἐλαυνομένου σιδήρου μετάθεσιν· σηκώματα ἀκριβῶς ἴσα, καὶ ῥοπὰς ἰσοπάτους τῶν ῥαιστήρων, πρὸς ἑαυτὸν ἀπηλλαγμένους. καὶ ἀπό τινος ἑνὸς παςάλου, διαγωνίως ἐμπεπηγότος τοῖς τοίχοις, ἵνα μὴ καὶ τούτου διαφορά τις ὑποφαίνηται· ἢ ὅλως ὑπονοήσαι παςάλων ἰδιαζόντων διαλλαγὴν, ἀπαρτήσας τέσσαρας χορδὰς ὁμοίους, καὶ ἰσοκώλους, ἰσοπαχεῖς τε καὶ ἰσοβόρους, ἑκάστην ἀφ' ἑκάστης ἐξήρτησεν, ὁλκὴν προσδήσας ἐκ τῶν κάτωθεν μέρους. τὰς δὲ μήκη τῶν χορδῶν μηχανησάμενος ἐκ παντὸς ἰσαίτους, εἶτα κρούων ἀνὰ δύο ἅμα χορδὰς ἐναλλὰξ συμφωνίας ἐλάμβανε τὰς προλεχθείσας, ἄλλην ἐν ἄλλῃ συζυγίᾳ. τὴν μὲν γὰρ ὑπὸ τοῦ μείζονος ἐξαρτήμα τεινομένην πρὸς τὴν ὑπὸ τοῦ μικροπέ-

nos inter se reddentes, una tantum combinatione excepta. In illis verò cognovit & dia pason, & dia pente, & dia tessaron concentum; at sonum, qui inter dia tessaron erat & dia pente, dissonum quidem per se existere videbat, sed completivum aliàs illius, qui inter ipsos esset maximus. Ingressus itaque, tanquam à Deo patefacta sibi intentione, in fabricam; ibique multis experimentis inventâ sonitus differentiâ, ex illis quæ in malleis erant gravitatibus; non verò à ferientium viribus; nec à malleorum figuris; nec à ferri, quod tundebatur, transpositione; cum pondera accurate excepissèt, & libramenta malleorum quam maxime æqualia; intra se est conversus, atque ab uno aliquo palo, qui ab uno angulo ad alterum muris esset impactus; ne & hinc differentia aliqua suboriri videretur, aut certe suspicionem adferret palorum, propriam naturam habentium, commutatio; adpendens quatuor chordas ejusdem materiæ existentes, tum æque longas, & æque crassas, atque æque graves, singulis singula pondera adpendit, alligata ex inferiori parte. Cumque ita chordarum longitudines omnino æquales effeciset, pulsans deinde binas atque binas simul chordas alternatim, consonantias inveniebat ante dictas, aliam in alia combinatione. Namque à maximo pondere tensam, ad eam quæ à mini-

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mo, dia paſon ſonantem deprehendit. Erat autem illa duodecim quarundam librarum; hæc verò, ſex. Atque ita in dupla ratione conſtituebat dia paſon conſonantiam, quam & ipſæ gravitates oſtendebant. Rurſus maximam ad juxta minimam, quæ octo librarum exiſtebat, dia pente conſonantem invénit. unde hanc in ratione ſeſquialtera conſtituit, in qua & ipſæ inter ſe erant libræ: Ad eam rurſus, quæ hanc ſequitur; quod ad gravitatem, iſtâ minorem, ſed reliquis majorem, quæ novem pondo eſſet; ipſam dia teſſaron, proportionaliter ipſis gravitatibus. Atque hanc ſuperquartam contrà deprehendebat, cum naturâ eadem ſeſquialtera eſſet minimæ. Quippe novem ad ſex ita habent, quemadmodum quæ juxta minimam eſt, octo libras habens, ad eam quidem, quæ ſex habet, in ratione erat ſuperquarta; ſed ad eam, quæ duodecim, in ſeſquialtera. Quod itaque eſt inter dia pente & dia teſſaron, hoc eſt, quo dia pente conſonantia ſuperat dia teſſaron, confirmatum eſt in ſuperoctava eſſe ratione, in qua novem ad octo. Porro dia paſon conſonantia utrorumque ſyſtema dicitur: ſeu ipſius dia pente, & dia teſſaron in conjunctione; ut dupla ratio ſyſtema eſt ſeſquialteræ & ſupertertiæ, in his numeris: duodecim, octo, ſex. aut contrà, ipſius dia teſſaron & dia pente; ut duplum, ſupertertii

λα, διὰ πασῶν φθεγγομένῳ καταλάμβανεν. ἓν δὲ ἡ μὲν, δώδεκά πνων ἑλκῶν· ἡ δὲ, ἕξ. ἐν διπλασίῳ δὴ λόγῳ ἀπέφαινε τὴν διὰ πασῶν, ὅπερ καὶ αὐτὰ βάρη ὑπέφαινε. τὴν δὲ αὖ μεγίστην πρὸς τὴν σμικροτάτην, ὅσων ὀκτὼ ὀλκῶν, διὰ πέντε συμφωνοῦσαν, ἔνθεν παύτην ἀπέφαινεν ἐν ἡμιολίῳ λόγῳ, ἐν ᾧπερ καὶ αἱ ὀλκαὶ ὑπῆρχον πρὸς ἀλλήλας. πρὸς δὲ τὴν μεθ' ἑαυτὴν μὲν τῷ βάρει· τῶν δὲ λοιπῶν μείζονα ἐννέα σταθμῶν ὑπάρχουσαν τὴν διὰ πασάρων, ἀναλόγως τοῖς βάρεσι. καὶ παύτην δὴ ὑπέτριτον ἄνισος καπλαμβάνει, ἡμιολίαν τὴν αὐτὴν φύσει ὑπάρχουσαν τῆς μικροτάτης. τὰ γὰρ ἐννέα πρὸς τὰ ἓξ ὅτως ἔχει, ὅνπερ τρόπον ἡ διὰ τὴν μικρὰν ἡ ὀκτὼ, πρὸς μὲν τὴν ἓξ ἔχουσαν, ὑπεμέτρῳ λόγῳ ἐστί· πρὸς δὲ τὴν τὰ δώδεκα, ἐν ἡμιολίῳ. τὸ ἄρα μεταξὺ τῆς διὰ πέντε καὶ τῆς διὰ πασάρων, τετάρτη, ᾧ ὑπερέχει ἡ διὰ πέντε τ διὰ πασάρων, ἐβεβαιῶτο ἐν ἐπογδόῳ λόγῳ ὑπάρχειν, ἐν ᾧπερ τὰ ἐννέα πρὸς τὰ ὀκτώ. ἑκατέρων τε ἡ διὰ πασῶν σύστημα ἐλέγετο, ἤτοι τ διὰ πέντε καὶ διὰ πασάρων ἐν συναφῇ· ὡς ὁ διπλάσιος λόγος ἡμιολίου τε καὶ ἐπιτρίτου. τοῖον, δώδεκα. ὀκτώ. ἕξ. ἢ ἀναστρόφως, τῆς διὰ πασάρων καὶ διὰ πέντε, ὡς τὸ διπλάσιον, ὑπεμέτρου τε καὶ

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καὶ ἡμιολίῳ, οἷον δώδεκα, ἐννέα, ἓξ, ἐν τάξει πιαύτῃ. τυλώσας καὶ τὴν χεῖρα, καὶ τῷ ἀκριβεῖ πρὸς τὰ ἐξαρτήματα, καὶ βεβαιώσας πρὸς αὐτὰ τὸν τῆς χορδῆς λόγον, μεπέβηκεν εὐμηχανῶς τῷ μὲν τῆς χορδῶν κοινῷ ἀποδοσιν, τῷ δὲ τοῦ διαγωνίου παρὰλληλον, εἰς τὸν τοῦ ὀργάνου βατῆρα, ὃν χορδότονον ὠνόμαζε· τῷ δὲ πρὸς τὴν ἀνάτασιν ἀναλόγως τῆς βάρεσιν, εἰς τῷ τῆς κολλάβων ἄνωθεν συμμετρον διεισροφῷ. ἐπιβάθρα τε πιάτῃ χρώμενος, καὶ οἷον ἀνεξαπατήτῳ γνώμονι εἰς ποικίλα ὄργανα λοιπὸν τῷ πεῖραν ἐξέτεινε. λεκίδων τε κρῦσιν, καὶ αὐλῶν, καὶ σύριγγας, καὶ μονόχορδα, καὶ τρίγωνα, καὶ τὰ παραπλήσια. καὶ σύμφωνον ἥυρισκεν ἐν ἅπασι καὶ ἀπερίληπτον τῷ δι' ἀριθμοῦ καταλῆψιν. ὀνομάζας δὲ ὑπάτην, τὸν τοῦ ἐξ ἀριθμοῦ κοινωνοῦντα φθόγγον. μέσην ἢ, τὸ ὀκτὼ, ἐπίτριτον αὐτῆ τυγχάνοντα. παραμέσην ἢ, τὸ ἐννέα, πέμπτῳ μέσης ὀξύτερον. καὶ δὴ καὶ ἐπόγδοον. νήτην ἢ, τὸ δώδεκα. καὶ τὰς μεταξύτητας κατὰ τὸ διατονικὸν γένος συναπληρώσας φθόγγοις ἀναλόγοις, ὅτως τὴν ὀκτάχορδον ἀριθμοῖς συμφώνοις ὑπέταξε. διαστασίῳ, ἡμιολίῳ, ἐπιτρίτῳ, καὶ τῇ τούτων διαφορᾷ, ἐπογδόῳ. τῷ δὲ προβάσιν ἀναγκῃ τινὶ

& ſeſquialteri, utputa in tali ordine: duodecim, novem, ſex. Cum itaque & manum, & auditum convertiſſet ad adpenſa pondera, atque ex iis confirmaſſet habitudinum rationem; machinatione facili tranſpoſuit; communem quidem chordarum adpenſionem, quæ ex palo erat ab angulo ad angulum infixo, in inſtrumenti metam, quam chordotonon, chordas tendentem lineam, nominabat: illam vero aliquantam diſtenſionem, proportionaliter ipſis gravitatibus, in clavulorum, qui ſupra erant, temperata converſione. Atque hac inceptione utens, & quaſi infallibili norma, in varia deinde inſtrumenta experimentum extendit; tum acetabulorum pulſationem, tum tibias, tum fiſtulas, & monochorda, & triangula, & adſimilia. In quibus omnibus conſonam invenit, & infallibilem, illam per numeros comprehenſionem. Cumque nominaſſet hypaten, numeri ſenarii participantem ſonum: meſen, octonarii, qui illius exiſtit ſupertertius: parameſen, novenarii participem, qui tono acutior eſt quam medius, tum quoque illius ſuperoctavus: neten denique, duodenarii: cumque interjecta ſpatia ſecundum genus diatonicum compleviſſet ſonis proportionalibus; ſic octo chordarum compaginationem conſonis numeris ſubjecit; duplo, ſeſquialtero, ſupertertio, atquè horum differentiæ, ſuperoctavo. Progreſſum vero, neceſſitate quadam naturali

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